Epoxy Anti-Corrosion Grey

Pure epoxy, interior · RAL 7037

RAL 7037 · Dusty grey

Product	Code • series	Colour reference	Document
Epoxy Anti-Corrosion Grey 环氧防腐灰	BL-I401 BL-I Industrial & Epoxy	RAL 7037 · Dusty grey	TDS BL-I401 Rev. 2026-10

Interior only — not part of the P-series. No UV resistance.

1. Description

The strongest adhesion and corrosion package in the range at 80–120 µm, for pipes, valves, tanks, enclosures and steel structures indoors. Epoxy chalks and yellows under UV and no stabiliser package fixes that. For outdoor corrosion duty use a polyester system, or the fluorocarbon P2000 for the top class.

2. Typical properties

Property	Value	Method
Chemistry	Pure epoxy, interior	—
Colour reference	RAL 7037 Dusty grey — approval on a sprayed panel against a physical chip	RAL
Finish and gloss	20–90 @60°	ASTM D523
Film build	80–120 µm	ISO 2808
Density	1.45–1.60 g/cm³	ISO 8130-2/3
Coverage	6.6 m²/kg at 100 % film · 6.2 m²/kg planning value at 95 % utilisation (with reclaim)	calculated
Cure schedule	180–200 °C × 10–15 min (object temperature)	object temp., not oven set point
Particle size D50	30–45 µm	ISO 8130-1 / -13
Stoving loss	≤ 1–2 %	ISO 8130-7
Storage stability	1 month at 35–40 °C	ISO 8130-8

3. Performance data

Test	Method	Typical result
1. Adhesion, cross-cut	ISO 2409 / ASTM D3359	0–1 (100/100 on phosphated steel)
2. Impact, direct and reverse	ASTM D2794	50 kg·cm, no cracking
3. Pencil hardness	ASTM D3363	2H
4. Gloss 60°	ASTM D523	20–90 per specification
5. Colour difference	ASTM D2244	ΔE ≤ 1.5 same batch
6. Neutral salt spray	ASTM B117 / ISO 9227	1,000 h at 100–120 µm
7. Chemical resistance	ASTM D1308	Good against alkali, dilute acid and solvents — state the medium and temperature
8. UV / exterior	—	Not applicable: chalks and yellows outdoors

Measured to the method listed, on production batches or the third-party report held for this system. Not a certificate: acceptance values are fixed in the trial report and the purchase specification.

4. Application and cure

01 Steel pre-treatment

Degrease → zinc phosphate → rinse → DI rinse → dry. Blast to Sa 2½ for immersion or C4/C5 duty; a phosphate alone is not enough for immersion.

02 Application

Corona 60–90 kV, 150–250 mm, 80–120 µm. Edges draw down: on cut edges and welds specify an edge-retentive grade or accept a thinner film there.

03 Cure

Object temperature 180–200 °C for 10–15 min. Epoxy is unforgiving of undercure — the salt-spray failure appears months later, not on the day.

04 Storage of coated parts

Fully cured before stacking. Parts stacked warm take print marks that cannot be buffed out of an epoxy film.

5. Packaging, storage and ordering

1. Standard pack: 20 kg net, polyethylene-lined carton. Other pack sizes on request.
2. Store below 30 °C, dry, out of direct sunlight, in the unopened pack. Keep away from ovens and hot lines.
3. Shelf life: 12 months from the production date in the original pack. Storage stability is verified at 35–40 °C for one month (ISO 8130-8).
4. A batch test report accompanies every delivery; a retained sample is kept for 12 months.
5. Sample in 3 days; production lead time confirmed per colour and pack size at order.
6. Colour match ΔE ≤ 1.5 within a batch. Cross-batch tolerance is agreed in the purchase specification.
7. Ask for: SDS · third-party test report · batch test report · physical colour chip.

6. Health and safety

1. Not classified as dangerous goods for transport. The SDS is available for every product — ask for it with the TDS.
2. The powder is a combustible dust. Earth the workpiece, the gun and the booth; no open flame or hot work in the booth; extract booth air; avoid dust clouds when emptying or cleaning.
3. Wear a P2 particulate mask, goggles and gloves when handling dry powder or cleaning the booth.
4. Cure volatiles: extract the oven and let parts cool before handling. Do not enter the oven zone without ventilation.

7. Also in this range

Code	Product	Colour	Film build	Cure
BL-M102	Bonded Gold Sand	RAL 1036	70–90 µm	200 °C × 10 min
BL-M103	Bonded Mirror Silver	RAL 9006	60–80 µm + clear	180–200 °C × 10–15 min
BL-A201	Outdoor Sand Grey	RAL 7035	60–80 µm	200 °C × 10 min

8. Document control

Sheet	Revision	Issued	Supersedes	Languages
BL-I401 TDS	2026-10	2026-10	None — first issue	English + 中文 (4 pp.)

9. Limits of this document

What this product is not for:

— Interior only. Outdoors it chalks within one to two years and the colour goes chalky white.

— Not for façade colour matching: epoxy shades are darker and less batch-stable than polyester.

— Food-contact or potable-water service: ask before quoting. We will say whether the report exists rather than assume it does.

1. Values in this sheet are typical for the product and were measured to the methods listed. They are not a certificate and not a warranty of fitness for a particular purpose. Acceptance values are fixed in the trial report and the purchase specification.
2. Colour: RAL references only. RAL does not define sRGB values, so the swatch in this sheet is a digital approximation — approval is made on a sprayed panel against a physical RAL chip.
3. No AAMA 2603 / 2604 / 2605, QUALICOAT, GSB or UL product certificate is held. We test to those methods and can supply third-party reports and in-house batch reports; if a tender requires the certificate itself, we will say plainly that we do not have it.
4. The 4,000 h salt spray and 10-year South Florida results shown elsewhere on our site belong to the fluorocarbon (PVDF) system, not to this powder product.

Issued by Guangdong Baolilai Industrial Co., Ltd. · jerry888.xu@gmail.com · +86 188 2310 0890
Ask for: SDS · third-party test report · batch report · physical colour chip · sample in 3 days

环氧防腐灰

纯环氧，室内型 · RAL 7037



产品	编号 • 系列	参照颜色	文件
Epoxy Anti-Corrosion Grey 环氧防腐灰	BL-I401 BL-I 工业防腐环氧	RAL 7037 · 灰土灰	TDS BL-I401 Rev. 2026-10

仅限室内——不属于 P 系列。不耐紫外。

1. 产品描述

全系最强的附着与防腐组合，膜厚 80–120 µm，用于室内的管道、阀门、储罐、机柜与钢结构。环氧在紫外下必然粉化黄变，加什么稳定剂都解决不了。户外防腐请用聚酯体系；最高等级用氟碳 P2000。

2. 典型性能

项目	数值	方法
体系	纯环氧，室内型	—
参照颜色	RAL 7037 灰土灰——以喷板对照实物色卡确认	RAL
表面与光泽	20–90（60°）	ASTM D523
膜厚	80–120 µm	ISO 2808
比重	1.45–1.60 g/cm³	ISO 8130-2/3
覆盖率	100% 成膜 6.6 m²/kg · 按 95% 利用率（含回收）规划值 6.2 m²/kg	计算值
固化条件	180–200 °C × 10–15 min（工件温度）	测工件温度，非炉膛设定
粒径 D50	30–45 µm	ISO 8130-1 / -13
烘烤失重	≤ 1–2 %	ISO 8130-7
贮存稳定性	35–40 °C 下 1 个月	ISO 8130-8

3. 性能数据

检测项目	方法	典型结果
1. 划格附着力	ISO 2409 / ASTM D3359	0–1（磷化钢件可达 100/100）
2. 正反冲击	ASTM D2794	50 kg·cm 不裂
3. 铅笔硬度	ASTM D3363	2H
4. 光泽 60°	ASTM D523	按规格 20–90
5. 色差	ASTM D2244	同批 ΔE ≤ 1.5
6. 中性盐雾	ASTM B117 / ISO 9227	膜厚 100–120 µm 达 1000 h
7. 耐化学性	ASTM D1308	耐碱、稀酸与溶剂良好——请说明介质与温度
8. 紫外 / 户外	—	不适用：户外粉化黄变

按所列方法在量产批次或本体系所持第三方报告上测定。不是证书：验收值以试粉报告与采购规格为准。

4. 施工与固化

01　钢件前处理

脱脂 → 锌系磷化 → 水洗 → 纯水洗 → 干燥。浸没或 C4/C5 工况喷砂至 Sa 2½；仅磷化不足以做浸没件。

02　喷涂

电晕 60–90 kV，枪距 150–250 mm，膜厚 80–120 µm。边角上膜偏薄：切割边与焊缝处请指定“边角保持型”，否则接受该处膜厚偏低。

03　固化

工件温度 180–200 °C × 10–15 min。环氧最怕固化不足——盐雾失效是几个月后才出现，不是当天。

04　涂装件存放

完全固化后再堆叠。带温堆叠会压出印痕，环氧涂层抛不掉。

5. 包装、贮存与订货

1. 标准包装：20 kg 净重，纸箱内衬聚乙烯袋。其他包装规格可订。
2. 30 °C 以下、干燥、避免阳光直射，保持原包装未开封。远离烘道与高温工位。
3. 保质期：原包装自生产日起 12 个月。贮存稳定性按 ISO 8130-8 于 35–40 °C 验证一个月。
4. 每批附批次检测报告；留样保存 12 个月。
5. 3 天出样；生产交期按颜色与包装规格在订单时确认。
6. 同批色差 ΔE ≤ 1.5；跨批公差在采购规格中约定。
7. 可索取：SDS · 第三方检测报告 · 批次检测报告 · 实物色卡。

6. 健康与安全

1. 不属于运输危险品。每个产品均提供 SDS（安全数据表），可与 TDS 一并索取。
2. 粉末为可燃性粉尘。工件、喷枪与喷房必须接地；喷房内禁止明火与动火作业；喷房须抽风；倒粉与清房时避免扬尘。
3. 处理干粉或清理喷房时，佩戴 P2 防尘口罩、护目镜与手套。
4. 固化挥发物：烘道须抽风；工件冷却后再搬运。无通风不得进入烘道区域。

7. 同系列其他产品

编号	产品	颜色	膜厚	固化
BL-M102	邦定金砂	RAL 1036	70–90 µm	200 °C × 10 min
BL-M103	邦定镜面银	RAL 9006	60–80 µm + clear	180–200 °C × 10–15 min
BL-A201	户外砂纹灰	RAL 7035	60–80 µm	200 °C × 10 min

8. 文件控制

文件	版本	签发	替代版本	语言
BL-I401 TDS	2026-10	2026-10	无——首版	英文 + 中文（共 4 页）

9. 本文件的边界

这款产品不适合用在：

— 仅限室内。户外一到两年即粉化，颜色变为灰白。

— 不用于幕墙配色：环氧色相比聚酯更深、批间稳定性更差。

— 食品接触或饮用水工况：报价前先问。我们会直接说报告有没有，不假设。

1. 本表数值为该产品的典型值，按所列方法测定。它们不是证书，也不构成对特定用途适用性的保证。验收值以试粉报告与采购规格为准。
2. 颜色：仅作 RAL 参照。RAL 未定义 sRGB 数值，本表色块为数字近似——以喷涂样板对照实物 RAL 色卡确认。
3. 我们不持有 AAMA 2603 / 2604 / 2605、QUALICOAT、GSB 或 UL 产品认证证书。我们按这些方法检测，可提供第三方报告与厂内批次报告；若招标硬性要求证书本身，我们会直接说明没有。
4. 我们网站上 4000 小时盐雾与 10 年南佛罗里达曝晒的结果属于氟碳（PVDF）体系，不属于本粉末产品。

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可索取：SDS · 第三方检测报告 · 批次报告 · 实物色卡 · 3 天出样